

Approximating Irrational Numbers

MA.8.NSO.1.1

I can tell a rational number from an irrational one, and approximate a root to the nearest tenth.

Lesson 1 asked what the side of an area-8 square was.

Today it gets a name and a value.

Warm-Up

On your own. Four minutes.

Same and different

What is the same about these two, and what is different?

$$\sqrt{49}$$

$$\sqrt{48}$$

Notes

I. A rational number's decimal stops or repeats

- A. rational number — a number that can be written as a ratio of two integers
- B. **As a decimal it either STOPS or REPEATS.**
- C. 0.375, -8 , $13/3$, $0.666\ldots$ are all rational.

Notes

II. An irrational number's decimal does neither

- A. irrational number — not a ratio of two integers
- B. Its decimal never stops and never repeats.
- C. π is irrational. 3.14 and $22/7$ are approximations — neither of them IS π .

Notes

III. The radicand decides, not the radical sign

The root	Rational or irrational?	Why
$\sqrt{25} = 5$	rational	25 is a perfect square
$\sqrt{18}$	irrational	18 is not a perfect square
$\sqrt[3]{27} = 3$	rational	27 is a perfect cube
$\sqrt[3]{30}$	irrational	30 is not a perfect cube

Notes

IV. Approximating to the nearest tenth — four steps

- A. **Find the two perfect squares (or cubes) the radicand sits between.**
- B. **That gives the two consecutive integers the root sits between.**
- C. Compare the two distances to decide which of the two it is nearer.
- D. **Test tenths. Square (or cube) them and see which lands closest.**

Example 1

$\sqrt{13}$. Steps 1 and 2, out loud.

A. Step 1: $9 < 13 < 16$

B. Step 2: $3 < \sqrt{13} < 4$

Which of the two integers is it nearer, and how do you know?

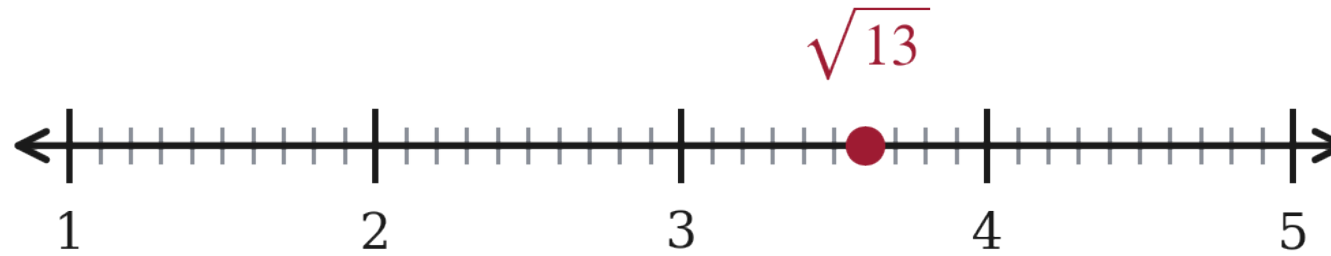
Example 1

Step 3, then test tenths.

- A. Step 3: $13 - 9 = 4$ and $16 - 13 = 3$, so it is nearer 4.
- B. **Step 4:** $3.6 \times 3.6 = 12.96$ $3.7 \times 3.7 = 13.69$
- C. 12.96 misses by 0.04. 13.69 misses by 0.69.

Example 1

$$\sqrt{13} \approx 3.6.$$



Example 2

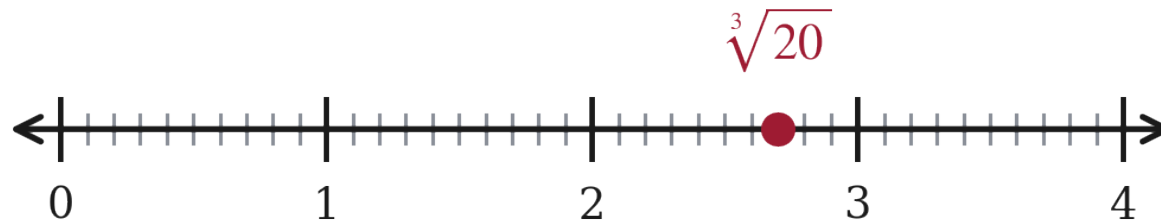
$\sqrt[3]{20}$. Same four steps. What changes?

Nothing on paper yet. Say which list you will use, and why.

Example 2

$$\sqrt[3]{20} \approx 2.7.$$

- A. Step 1: $8 < 20 < 27$ two perfect CUBES
- B. Step 2: $2 < \sqrt[3]{20} < 3$
- C. Step 3: $20 - 8 = 12$ and $27 - 20 = 7$, so it is nearer 3.
- D. **Step 4: $2.7^3 = 19.683$ $2.8^3 = 21.952$**



Example 3

Priya and her calculator disagree.

Priya rounds first	The calculator does not
$\sqrt{6} \approx 2.4$	$\sqrt{6} = 2.449489\dots$
$2 \times 2.4 = 4.8$	$2 \times 2.449489\dots = 4.898979\dots$

One of them is wrong. Which, and why?

Example 3

Priya is.

$2\sqrt{6} = 4.898979\dots$, which is 4.9 to the nearest tenth.

- A. Rounding 2.449 down to 2.4 throws away 0.049.
- B. **Doubling throws away 0.098 — almost a whole tenth, which is the whole answer.**

Example 3

So: round last.

Using 2.45 instead: $2 \times 2.45 = 4.90$ ✓

Whiteboards • Question 1 of 9

Boards up on three.

$$\sqrt{49}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards • Question 1 of 9

Answer.

$$\sqrt{49}$$

49 is a perfect square

Answer: rational, 7

Whiteboards • Question 2 of 9

Boards up on three.

$$\sqrt{48}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards · Question 2 of 9

Answer.

$$\sqrt{48}$$

$36 < 48 < 49$, so it is just under 7

Answer: irrational, ≈ 6.9

Whiteboards • Question 3 of 9

Boards up on three.

$$\sqrt[3]{48}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards · Question 3 of 9

Answer.

$$\sqrt[3]{48}$$

$27 < 48 < 64$, so it is between 3 and 4

Answer: irrational, ≈ 3.6

Whiteboards • Question 4 of 9

Boards up on three.

$$\sqrt[3]{64}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards • Question 4 of 9

Answer.

$$\sqrt[3]{64}$$

64 is a perfect cube

Answer: rational, 4

Whiteboards • Question 5 of 9

Boards up on three.

$$\sqrt{64}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards • Question 5 of 9

Answer.

$$\sqrt{64}$$

64 is a perfect square TOO

Answer: rational, 8

Whiteboards • Question 6 of 9

Boards up on three.

$$\sqrt{8}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards · Question 6 of 9

Answer.

$$\sqrt{8}$$

$4 < 8 < 9$, so it is just under 3

Answer: irrational, ≈ 2.8

Whiteboards • Question 7 of 9

Boards up on three.

π

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards • Question 7 of 9

Answer.

π

3.14159265... — it never repeats and never stops

Answer: irrational, ≈ 3.14

Whiteboards • Question 8 of 9

Boards up on three.

$$0.\overline{6}$$

Rational or irrational? Give its value.

If it is irrational, the nearest tenth is enough.

Whiteboards · Question 8 of 9

Answer.

$$0.\overline{6}$$

$0.666... = 2/3$, a ratio of two integers

Answer: rational, = 2/3

Whiteboards • Question 9 of 9

Take your time. Boards up when you have written it.

Marcus says any decimal that goes on forever is irrational. Is he right?

Write your answer in sentences.

This one is written work. Give an example that proves your answer.

Whiteboards • Question 9 of 9

Answer.

Marcus says any decimal that goes on forever is irrational. Is he right?

0.666... goes on forever and equals $\frac{2}{3}$

Answer: No — it has to go on WITHOUT REPEATING

IXL

Last five minutes.

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Whatever you do not finish is tonight's practice.
4. The skills for today are posted in Google Classroom.